

# PE-EC801A — Antennas and Propagation

Semester VIII • 3L:0T:0P • 3 Credits

## Module 4 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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### Contents

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|          |                                        |          |
|----------|----------------------------------------|----------|
| <b>1</b> | <b>Yagi-Uda Antenna</b>                | <b>2</b> |
| 1.1      | Structure and Elements                 | 2        |
| 1.2      | Characteristics                        | 2        |
| <b>2</b> | <b>Log-Periodic Antennas</b>           | <b>3</b> |
| 2.1      | Geometric Scaling Laws                 | 3        |
| 2.2      | Principle of Operation (Active Region) | 3        |
| <b>3</b> | <b>Frequency Independent Antennas</b>  | <b>4</b> |
| 3.1      | Rumsey’s Principle                     | 4        |
| 3.2      | Examples                               | 4        |
| <b>4</b> | <b>Broadcast Antennas</b>              | <b>4</b> |
| 4.1      | Turnstile Antenna                      | 4        |
| 4.2      | Vertical Mast Radiators                | 5        |

## 1. Yagi-Uda Antenna

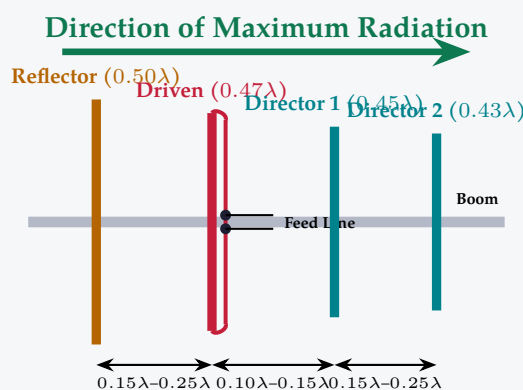
### Definition of Yagi-Uda Antenna

The **Yagi-Uda antenna** is a highly directional, narrow-band antenna consisting of a single driven element (usually a half-wave folded dipole) and one or more parasitic elements (a reflector behind it, and directors in front of it) aligned along a central supporting boom.

### 1.1 Structure and Elements

The parasitic elements are not connected to the feed line. They receive energy through mutual electromagnetic coupling from the driven element and re-radiate it.

#### Yagi-Uda Antenna Layout



- 1. Driven Element:** Typically a folded dipole of resonant length  $0.47\lambda - 0.48\lambda$ . The folded dipole structure increases the input impedance from  $73\Omega$  to around  $300\Omega$ , facilitating impedance matching with standard twin-lead transmission lines.
- 2. Reflector:** Placed behind the driven element. It is approximately 5% longer than the driven element ( $\approx 0.50\lambda$ ) and spaced  $0.15\lambda - 0.25\lambda$  away. Because it is longer, it is inductive. It redirects the wave in the forward direction.
- 3. Directors:** Placed in front of the driven element in the direction of the main beam. They are 5% shorter than the driven element ( $\approx 0.40\lambda - 0.45\lambda$ ) and spaced  $0.10\lambda - 0.30\lambda$  apart. Because they are shorter, they are capacitive, directing wave propagation forward.

### 1.2 Characteristics

- **Directivity/Gain:** Typically 7–15 dB depending on the number of director elements.
- **Bandwidth:** Narrow (typically 2% to 10%).
- **Front-to-Back Ratio:** High, meaning it effectively rejects noise from the back.

## 2. Log-Periodic Antennas

### Definition of Log-Periodic Antenna

A **Log-Periodic Dipole Array (LPDA)** is a multi-element directional antenna designed to operate over an extremely wide bandwidth. Its electrical characteristics (gain, radiation pattern, input impedance) repeat periodically with the logarithm of the frequency.

## 2.1 Geometric Scaling Laws

The physical dimensions of the LPDA elements scale according to a geometric progression. Let  $L_n$  be the length of the  $n$ -th dipole element,  $d_n$  be the spacing between the  $n$ -th and  $(n + 1)$ -th elements, and  $R_n$  be the distance from the apex to the  $n$ -th element.

### LPDA Scaling Ratios

The geometric scale factor  $\tau$  and spacing factor  $\sigma$  are defined as:

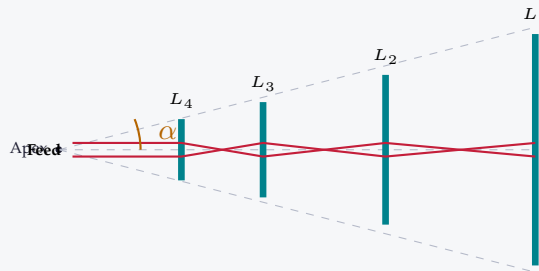
$$\tau = \frac{L_{n+1}}{L_n} = \frac{R_{n+1}}{R_n} = \frac{d_{n+1}}{d_n} \quad (0 < \tau < 1)$$

$$\sigma = \frac{d_n}{2L_n}$$

The apex angle  $2\alpha$  of the structure is related to these ratios by:

$$\tan \alpha = \frac{L_n - L_{n+1}}{2R_n} = \frac{1 - \tau}{4\sigma}$$

### Log-Periodic Dipole Array (LPDA)



## 2.2 Principle of Operation (Active Region)

The feed line is cross-phased between adjacent elements (a  $180^\circ$  phase twist). This configuration ensures that at any given frequency:

- **Transmission Region:** Elements that are too short ( $L \ll \lambda/2$ ) present a high capacitive reactance and do not draw much current; they act as a transmission line.
- **Active Region:** Elements whose length is close to half a wavelength ( $L \approx \lambda/2$ ) are resonant. They draw the majority of the current and radiate.
- **Reflective Region:** Elements that are too long ( $L \gg \lambda/2$ ) are highly inductive and act as reflectors.

As the operating frequency increases, the active region shifts toward the smaller elements at the apex, maintaining constant beamwidth and input impedance.

### 3. Frequency Independent Antennas

#### 3.1 Rumsey's Principle

Rumsey's Principle states that an antenna will be frequency-independent if its physical shape is defined entirely by angles, with no characteristic scale length. If the dimensions are scaled, the antenna looks identical except for a rotation.

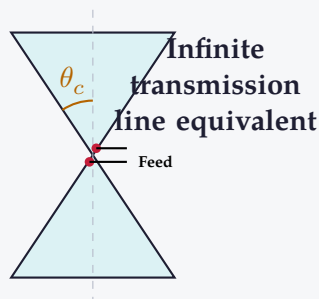
#### 3.2 Examples

- **Biconical Antenna:** Formed by two co-axial metallic cones with their vertices facing each other. It behaves as a broadband version of a dipole.
- **Equiangular Spiral Antenna:** The boundary curves are defined in polar coordinates by:

$$r = r_0 e^{a\phi}$$

Because the shape is defined purely by exponential growth and angles, it has infinite theoretical bandwidth (practically limited by feed-point dimensions at high frequencies and outer boundaries at low frequencies).

#### Biconical Antenna Geometry



### 4. Broadcast Antennas

Broadcast applications require omnidirectional radiation patterns in the horizontal plane to cover a  $360^\circ$  service area.

#### 4.1 Turnstile Antenna

A **turnstile antenna** consists of two identical horizontal half-wave dipoles placed perpendicular to each other ( $90^\circ$  spatial separation) and excited with a  $90^\circ$  phase difference.

- The E-field components in the horizontal plane are:

$$E_1 = E_0 \cos \phi, \quad E_2 = E_0 \sin \phi e^{j\pi/2} = jE_0 \sin \phi$$

- The total field is:

$$E_{total} = E_1 + E_2 = E_0(\cos \phi + j \sin \phi) = E_0 e^{j\phi}$$

- The magnitude  $|E_{total}| = E_0$  is constant, producing a perfectly circular (omnidirectional) radiation pattern in the horizontal plane with circular polarization.

## 4.2 Vertical Mast Radiators

For Medium Frequency (MF) AM radio broadcasting, vertical steel towers (mast radiators) of height  $H \approx \lambda/4$  or  $H \approx 0.583\lambda$  (the anti-fading height, which minimizes sky-wave interference near the ground) are used. The ground acts as a reflector, forming a monopole equivalent.

## Quick Revision — Module 4

Broadband and Broadcast Antennas | PE-EC801A

### Key Formulas and Metrics at a Glance

|                                  |                                                                              |
|----------------------------------|------------------------------------------------------------------------------|
| LPDA Scale Factor ( $\tau$ )     | $\tau = \frac{L_{n+1}}{L_n} = \frac{R_{n+1}}{R_n} = \frac{d_{n+1}}{d_n} < 1$ |
| LPDA Spacing Factor ( $\sigma$ ) | $\sigma = \frac{d_n}{2L_n}$                                                  |
| LPDA Apex Angle ( $\alpha$ )     | $\tan \alpha = \frac{1-\tau}{4\sigma}$                                       |
| Log-Spiral Geometry              | $r = r_0 e^{a\phi}$ (defined purely by angles)                               |
| Yagi Driven Element Length       | $L \approx 0.47\lambda - 0.48\lambda$ (usually folded dipole)                |
| Yagi Reflector Length            | $L \approx 0.50\lambda$ (inductive, reflects wave forward)                   |
| Yagi Director Length             | $L \approx 0.40\lambda - 0.45\lambda$ (capacitive, steers wave forward)      |

### One-Line Definitions

- **Parasitic Element:** An antenna element not electrically connected to the receiver/transmitter, excited via mutual coupling.
- **Folded Dipole:** A dipole antenna with its ends folded back and joined, providing a four-fold increase in input impedance.
- **Active Region:** The zone in a log-periodic array where elements are near resonance ( $\approx \lambda/2$ ) and actively radiating.
- **Rumsey's Principle:** The condition that an antenna shape defined only by angles is frequency-independent.
- **Turnstile Antenna:** An arrangement of two crossed dipoles fed out of phase to produce a circular, omnidirectional horizontal pattern.
- **Anti-Fading Mast:** A vertical tower radiator of height  $\approx 0.583\lambda$  that suppresses high-angle sky-wave radiation.

### Frequently Asked / Viva Questions

#### Q1. Explain the operation of parasitic elements in a Yagi-Uda antenna.

The reflector (longer than  $\lambda/2$ ) acts inductively, shifting the phase of its re-radiated field to add constructively with the driven element's field in the forward direction. The directors (shorter than  $\lambda/2$ ) act capacitively, shifting the phase to pull the beam forward.

#### Q2. Why is a folded dipole used as the driven element in a Yagi-Uda antenna?

Adding parasitic elements near a dipole severely lowers its input impedance (often down to  $< 15\Omega$ ), causing mismatch loss. A folded dipole has a high impedance ( $300\Omega$  in free space), which drops to a convenient  $50-75\Omega$  in the array,

## Frequently Asked / Viva Questions (contd.)

making it easy to match to coaxial cables.

**Q3. What is the difference between a broadband antenna and a frequency-independent antenna?**

A broadband antenna operates over a wide range (e.g., Yagi-Uda with 10% bandwidth). A frequency-independent antenna satisfies Rumsey's principle (angular geometry) and has a practically unlimited bandwidth (often 10 : 1 or more), as its active zone shifts continuously with frequency.

**Q4. How does a Log-Periodic Dipole Array (LPDA) achieve frequency independence?**

As frequency changes, the "active region" (elements near  $\lambda/2$  resonance) moves smoothly along the boom. The active elements scale geometrically, so the antenna's electrical size relative to the operating wavelength remains constant.

**Q5. What parameters define the geometry of an LPDA?**

The geometric scale factor  $\tau$  (ratio of adjacent lengths/spacings), the spacing factor  $\sigma$  (ratio of spacing to element length), and the apex angle  $\alpha$ .

**Q6. How does a Turnstile antenna produce an omnidirectional pattern in the horizontal plane?**

It uses two crossed horizontal dipoles fed in phase quadrature ( $90^\circ$  phase shift). The resulting fields  $E_0 \cos \phi$  and  $jE_0 \sin \phi$  combine vectorially to yield a constant amplitude  $|E| = E_0$  at all horizontal angles  $\phi$ .

**Q7. Why is the height of  $0.583\lambda$  chosen for AM broadcast mast radiators?**

At  $0.583\lambda$ , the antenna produces maximum ground-wave radiation while minimizing high-angle sky-wave radiation. This reduces interference between the ground wave and the ionospheric sky wave, preventing fading in the primary coverage zone.

## Yagi-Uda vs. Log-Periodic Dipole Array (LPDA)

| Parameter         | Yagi-Uda Antenna                              | Log-Periodic (LPDA)                             |
|-------------------|-----------------------------------------------|-------------------------------------------------|
| Bandwidth         | Narrow (typically 2%–10%)                     | Extremely Wide (10 : 1 ratio or more)           |
| Gain              | High (7–15 dB depending on elements)          | Moderate (6–10 dB)                              |
| Element Design    | Parasitic elements of specific lengths        | Driven elements of geometrically scaled lengths |
| Feeder Connection | Driven element only (others mutually coupled) | All elements connected to a cross-phased line   |